



# Guillaume Falmagne

Research Fellow and team leader in complex systems (TUD and TCD)

## Research experience

## Education

**Research Fellow and Principal Investigator**  
2025 – 2030 SOHAM centre, TU Dublin & Trinity College Dublin  
COOLNET project (€1 million grant):  
large-scale cooperation investigated with video games

**Postdoctoral research associate**  
2022 – 2025 High Meadows Environmental Institute, Princeton University  
Complex socio-ecological systems  
Supervisor: Simon LEVIN

Summer 2024

Complex Systems Summer School  
Santa Fe Institute (New Mexico)

4-week interdisciplinary course on complex behavior

**Postdoctoral researcher**  
Subatech (IMT Atlantique), Nantes  
Spring 2022 Partonic energy loss in the quark-gluon plasma  
Supervisor: François ARLEO

2018 – Dec 2021 The  $B_c^+$  meson in heavy ion collisions with the CMS detector;  
Partonic energy loss in the quark-gluon plasma  
supervisors: Raphaël GRANIER DE CASSAGNAC, François ARLEO

**PhD in particle physics**  
LLR, École Polytechnique (IP Paris)

Courses: Advanced QFT,  
CERN-Fermilab HCP school (2019),  
Int. School of QCD (LPT Orsay, 2018),  
models for the energy transition, plant seeds

2017–2018 **5-month research internship**  
feasibility study on PhD topic

**MSc (year 2) in High Energy Physics**  
École Polytechnique (Université Paris-Saclay)  
summa cum laude

end 2016 – jun 2017

Volunteering and cultural experience in South America & Eastern Europe

2015 – end 2016 **12-month research internship**, CERN, Geneva  
 $\Lambda_b^0$  production charge asymmetry measurements with the LHCb detector  
supervisor: Sascha STAHL

**4-month research internship (2015)**  
SLAC, Stanford University  
NNLO-PS matching & jet clustering in SHERPA  
supervisors: Lance DIXON, Stefan HOECHE

2013–2015

**2-month research internship (2014)**  
LLR, École Polytechnique  
Resolution unfolding of the CMS detector  
supervisor: Raphaël GRANIER DE CASSAGNAC

**BSc degree & MSc (year 1),  
Fundamental Physics**  
École Normale Supérieure Cachan (Paris-Saclay)  
& Université Pierre et Marie Curie  
magna cum laude

2011–2013

Classes préparatoires in Math & Physics  
Lycée Henri Poincaré (Nancy, France)

2011

Baccalauréat in Sciences, summa cum laude

✉ [guillaume.falmagne@tcd.ie](mailto:guillaume.falmagne@tcd.ie) • [sites.google.com/view/guillaumefalmagne/](https://sites.google.com/view/guillaumefalmagne/)

📞 0000-0002-6762-3937

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## Awards and fellowships

- 2024 Recipient of the **Prize of the Lopez-Loreta Foundation** for the CORESO/COOLNET project, €1 million grant over 5 years
- 2023 **Best poster (2nd place) of the Collective Intelligence Symposium**, Santa Fe Institute
- 2022 PhD thesis award of **Institut Polytechnique de Paris**
- 2022 PhD thesis award of **Groupeement de Recherche QCD**
- 2022 Accessit to the **Daniel Guinier PhD thesis award of Société Française de Physique**
- 2022 **CERN Senior Research Fellowship**, declined for HMEI Research Associate position, Princeton U.
- 2013 **Admission at ENS Cachan after national competitive exams**, 'normalien': full 4-year funding

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## Publications

### Complex systems

- 2026 **The phase space of symmetric  $2 \times 2$  games**, Guillaume Falmagne, submitted to *Int. Journal of Game Theory*, short version accepted to *SIAM News*, [SocArXiv tgh9y\\_v1](#),  Solo author (2025-26)
- 2026 **Inseparability as foundation: rethinking risk science for coupled, co-evolving systems**, Alexandre Dunant + 20 authors, white paper, submitted to *Risk Analysis*  
 Contribution to initial IIASA workshop discussions (Nov 2025) and writing (2025-26)
- 2026 **From risk to flourishing: Seeds for responding to a complex world**, Ryan McGranaghan + 8 authors, perspective piece, submitted to *Risk Analysis*  
 Contribution to inception, discussions and writing (2025-26)
- 2026 **From phase transitions to tipping points: Theory and empirics of critical transition phenomena**, Elizabeth Krueger + 21 authors, submitted to *Annual Review of Environment and Resources*  
 Contribution to inception and Physics review, main *Critical Transitions Workshops* organizer (2023-26)
- 2026 **Interpretable early warnings using machine learning in an online game-experiment**, Guillaume Falmagne, Anna B. Stephenson, Simon A. Levin, **PNAS (2026) 123**, [e2503493122](#),  Lead author (2022-25)

### Particle physics (theory)

- 2024 **Path-length dependence of parton and jet energy loss from universal scaling laws**, François Arleo, Guillaume Falmagne, **Proceedings of Science, ICHEP2024 (2025) 604**
- 2024 **Probing the path-length dependence of parton energy loss via scaling properties in heavy ion collisions**, François Arleo, Guillaume Falmagne, **PRD Letters 109 (2024), L051503**
- 2019 **Quenching of hadron spectra in XeXe and PbPb collisions at the LHC**, François Arleo, Guillaume Falmagne, **Proceedings of Science, HardProbes2018 (2019) 075**
- 2016 **Les Houches 2015: Physics at TeV Colliders Standard Model WG Report**, J.R. Andersen et al. (~100 authors), [arXiv:1605.04692](#),  Contribution to NNLO-PS matching in V-Higgs + MINLO procedure

### Particle physics (experimental)

Following the publication model of CMS, all papers are signed by all (>2000) authors.

- 2022 **Observation of the  $B_c^+$  Meson in Pb-Pb and pp Collisions at  $\sqrt{s_{NN}} = 5.02$  TeV and Measurement of its Nuclear Modification Factor**, CMS collaboration, **PRL 128, 252301**  
 Contact author and analysis lead (2018-2021)
- 2021 **Observation of a  $\Lambda_b^0 - \bar{\Lambda}_b^0$  production asymmetry in proton-proton collisions at  $\sqrt{s} = 7$  and 8 TeV**, LHCb collaboration, **JHEP10 (2021) 060**,  Major role in analysis (2015-17)
- 2019–2021 **Author of all (> 300) papers from the CMS Collaboration submitted in this period**  
 Direct contributions to:  
**Performance of CMS muon reconstruction from proton-proton to heavy ion collisions** (**JINST (2024) 19 P09012**),  
**Fragmentation of jets containing a  $J/\psi$  meson in PbPb and pp collisions at  $\sqrt{s_{NN}} = 5.02$  TeV** (**PLB(2021) 136842**)

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## Work in progress

### Active

- 2025-2026 **Adaptive AI Influence Shapes Human-AI Collective Belief Dynamics**, with Hao Cui and Taha Yasserli (SOHAM, TUD&TCD)
- 2022-2026 **A dynamical model for tipping cascades due to the moisture recycling network in the Amazon rainforest**, with Nico Wunderling (PIK)
- 2024-2026 **Evaluating the effect of network structure on community success using the r/place game-experiment**, with Anna Stephenson (Princeton) and Abraham Israeli (U. of Michigan)
- 2023-2026 **Understanding scaling laws and optimization principles of organizations using the r/place game-experiment**, with Anna Stephenson (Princeton) and Chris Kempes (Santa Fe Institute)
- 2023-2026 **The large-scale propagation of cooperation on multilayer networks**, with Giuseppe Ferro (Princeton), Woi Sok Oh (Princeton), and Emma Zajdela (Princeton)

### Paused

- 2024-2025 **Effect of timescales mismatch on large-scale social dilemmas**, with Saverio Perri (UC Riverside), Emma Zajdela (Princeton), Giulio Burgio (UVM)
- 2023-24 **Extracting uni-dimensional voter spectra from ranked choice voting ballots**, with Samuel S. Wang (Princeton), Keena Lipsitz (CUNY), Simon A. Levin (Princeton), and Annie Stephenson (Princeton)
- 2024 **Collective dynamics of singing pitch propagation and convergence**, SFI CSS School project with Galen Hall and Ritam Pal

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## Teaching experience

- Fall 2024 **Instructor of the *Programming for Biology* course**, EEB department, Princeton University
- 2023 – 2025 Private research mentor, CCIR
- 2018 – 2021 **Teaching assistant**, École Polytechnique (Institut Polytechnique de Paris)  
Optics, Waves, and Radiation (Bachelor 2<sup>nd</sup> year). Advanced Particle Physics (Master 1<sup>st</sup> year).
- 2020 **Research supervisor**, École Polytechnique  
MSc year 1 (2 months), BSc year 2 (2 months)
- 2014 – 2016 Private tutoring, LiveMentor  
Mathematics and physics for students in classes préparatoires.
- nov-dec 2014 Teaching internship in physics in high school, Lycée Frédéric Mistral, Fresnes (France)

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## Service

### Conference organization

- Expected Spring 2027 **Lead organizer, Gamification of Science Workshop**, COOLNET and SOHAM, TU Dublin and Trinity College Dublin
- Spring 2026 **Co-organizer, SOHAM Seminar**, SOHAM Centre at TUD and TCD, Dublin
- Feb 2025 **Lead organizer, 6th Critical Transitions Workshop**, including securing funding, [Princeton Center for Theoretical Science](#), USA
- Dec 2023 **Central organizer, 4th Critical Transitions Workshop**, [Potsdam Institute for Climate Impact Research \(PIK\)](#), Potsdam (Germany)

### Scientific outreach

- Nov 2023 **Two pictures exhibited in “Birdiversity”**, Princeton  
Communicating the science and challenges of bird diversity through art
- oct 2019 Volunteer at **CERN Open Days**, CMS site, Geneva  
Outreach on CMS activities, 80,000 visitors in total in a week-end
- apr 2019 Animation of the CERN stand at **Geek Touch Convention**, Lyon (France)  
Demonstrating CERN activities to a broad public

## Lab and school life

- 2023 Member of the **Climate Committee** of the EEB department  
D&I and general climate
- 2019–2022 **Representative** of students and postdocs and sustainable development leader at **LLR Lab Council**
- 2014–2015 Member of the **student association** of the ENS Cachan  
In charge of communication and partnerships
- nov 2013 Humanitarian association of ENS Cachan: social microcredit event (*The Rise*, Babyloan)

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## Scientific talks

- 2026, *forthcoming* **Interacting transitions workshop, keynote presentation**, Centre for Complex Systems Studies, Utrecht University, *Transitions in large-scale online systems*
- 2025 Cooperation and Collective Cognition Network (CoCCoN) workshop, HU Berlin, *COOLNET: A video game as a collective human behavior lab*
- 2025 **Sunbelt**, Social Networks, Paris, *CORES0/COOLNET: new tools for large-scale cooperation*
- 2025 French chapter of the **Complex Systems Society**, Paris, *Interpretable Early Warning System using Machine Learning in a Large-scale Game-experiment*
- 2025 French Conference on Complex Systems, FRCCS, Bordeaux
- 2025 CS2Italy (Computational Social Science), University of Trento
- 2024 “Many Minds” meeting, poster, Dpt of Psychology, Princeton University
- 2024 Workshop on spreading in social networks, DIMACS Rutgers University
- 2024 ERSI Critical Transitions Workshop, Stockholm Resilience Center
- 2024 CoCCoN (Cooperation and Collective Cognition Network) seminar, HU Berlin, online, *CORES0/COOLNET: COOperation in Large-scale NETworks*
- 2024 **European Geosciences Union (EGU) General Assembly**, Vienna (Austria), [online](#), *Interpretable Early Warning System using Machine Learning in a Large-scale Game-experiment*
- 2024 **International School and Conference on Network Science (NetSciX)**, Venice (Italy), *Tipping cascades in the Amazon rainforest due to the moisture recycling network*
- 2023 ERSI Critical Transitions Workshop, PIK, Berlin (Germany), *Trees, networks and games for large-scale cooperation*
- 2023 Seminar at Theoretical Ecology Tea, EEB, Princeton, *Structural aspects of large-scale cooperation*
- 2023 GdR QCD General Assembly, talk for 2022 PhD award, [online](#), *Collective phenomena and critical transitions: from plasma to complex socio-ecological systems*
- 2023 **Collective Behaviour Workshop**, [Isaac Newton Institute](#), Cambridge (UK), *Early warning signals and the structure and emergence of collaborations with a large-scale experiment*
- 2023 Seminar at Max Planck Institute for Human Development, Berlin (Germany), *Early warning signals and the structure of collaborations using a large-scale experiment on Reddit*
- 2023 Seminar at PIK (Potsdam Institute for Climate Impact Research, Germany), *Investigating early warning signals with a large-scale collaborative experiment on Reddit*
- 2023 **Collective Intelligence Symposium**, [poster](#), Santa Fe Institute (NM, USA), *Surveying early warning signals of transitions using a large-scale collaborative experiment*
- 2021 **CERN-LHC Seminar**, [online \(recording available\)](#), *First observation of the  $B_c^+$  meson in PbPb and pp collisions at 5.02 TeV at CMS*
- 2021 Rencontres QGP France, [Étretat](#) (France), *First observation of  $B_c^+$  meson production in PbPb and pp collisions with CMS*
- 2021 **9th Edition of Large Hadron Collider Physics Conference (LHCP)**, [online \(recording available\)](#), *Exotic quarkonia production in heavy ion collisions:  $X(3872)$  and  $B_c^+$*
- 2021 **19th International Conference on Strangeness in Quark Matter (SQM)**, [online](#), *First measurement of the  $B_c^+$  meson nuclear modification factor in PbPb collisions with CMS*
- 📧 Published proceedings: [EPJ Web of Conferences 259](#), 12011 (2022)

- 2021 **14th International Workshop on Heavy Quarkonium (QWG)**, [online](#),  
*Quarkonium production studies in nuclear collisions at CMS*
- 2020 Journées CMS-France, [online](#), *B<sub>c</sub> production: Towards a first observation in heavy ions collisions*
- 2019 Journées de Rencontre des Jeunes Chercheurs, [Moulin-Mer](#) (France),  
*Probing the quark-gluon plasma with the B<sub>c</sub> meson in CMS*  
 📖 **Published proceedings:** C. Armand et al., [JRJC 2019 Book of Proceedings](#), p.88
- 2019 **GdR Intensity Frontier Workshop**, [Sommières](#) (France),  
*B<sub>c</sub> meson production in pp and PbPb collisions with CMS*, and plenary talk (*B<sub>c</sub> session summary*)
- 2019 QGP France, [Étretat](#) (France),  
*Modification of B<sub>u</sub><sup>+</sup>, B<sub>s</sub><sup>0</sup> and B<sub>c</sub><sup>+</sup> mesons in PbPb collisions with the CMS detector*
- 2018 **Int. Conf. on Hard & EM Probes of High-Energy Nuclear Collisions**, [Aix-les-Bains](#) (France),  
*B<sub>s</sub><sup>0</sup> and B<sup>+</sup> meson nuclear modification factors in PbPb collisions at 5.02 TeV with CMS detector*  
 📖 **Published proceedings:** G. Falmagne for the CMS Collaboration, Proc. of Hard Probes 2018, [PoS 143](#)

## Languages

French, native/fluent  
 English  
 German Reading, writing, conversation  
 Spanish Reading, conversation

## Non-professional activities

Rugby 10+ years in clubs and a university team  
 Scuba-diving CMAS two star diver  
 Running Typical performance: 10km in 40min  
 Music Flute (classical formation, university bands), classical singing (multiple choirs)  
 Photography [Travel blog](#), pictures exhibited in “[Birdiversity](#)” (Princeton) and in “[A World Of Stations](#)” (Paris),  
[instagram](#)